Lecture 14

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Foreign Exchange Market Conventions

- Foreign exchange rates are quoted as prices of one currency in terms of another.
- Let: S_0= Spot exchange rate
- Typically expressed as domestic currency per unit of foreign currency
- Example: S_0 = 1.10 USD/EUR
  - means: 1 euro costs 1.10 U.S. dollars.
- Direct vs Indirect Quotes
  - Direct quote (U.S. perspective): USD per foreign currency
  - Indirect quote: Foreign currency per USD
- Bid–ask spreads reflect dealer compensation and market liquidity.
- Currency pairs are always quoted relative to a base currency.

Spot and Forward Exchange Rates

- The spot rate S_0 is the price for immediate exchange.
- The forward rate F_0 is agreed today for exchange at future time T.
- Forward contracts eliminate future exchange rate uncertainty.
- The forward rate is determined by no-arbitrage.
- It is not a forecast of future spot.

Covered Interest Rate Parity (CIP)

- Suppose:
  - r_d = domestic interest rate
  - r_f = foreign interest rate
  - S_0 = spot rate
  - F_0 = forward rate
- No-arbitrage implies: $F_0 = S_0 e^{(r_d - r_f)T}$
- Under simple compounding: $F_0 = S_0 \frac{1 + r_d T}{1 + r_f T}$
- This is covered interest parity (CIP).

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Arbitrage Logic Behind CIP

- Two strategies must yield the same domestic currency value at time T:
  1. Invest domestically at r_d
  2. Convert to foreign currency at S_0, invest at r_f,lock in forward rate F_0
- No-arbitrage requires equality.
- If violated:
  - Borrow in low-return currency
  - Invest in high-return currency
  - Hedge FX risk
- CIP is enforced by capital flows.

Interpretation of Forward Premium

- Define forward premium: $\frac{F_0 - S_0}{S_0}$
- From CIP: $\frac{F_0 - S_0}{S_0} \approx r_d - r_f$
- If domestic rates exceed foreign rates:
  - Domestic currency trades at forward discount
  - Foreign currency trades at forward premium
- Forward prices reflect interest rate differentials.

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Numerical Example: Covered Interest Parity

- Suppose:

$$S_0 = 1.10 \text{ USD/EUR}$$

$$r_d = 5\%,\ r_f = 3\%,\ T=1$$

- Then:

$$F_0 = 1.10e^{(0.05-0.03)}$$

$$F_0 \approx 1.10e^{0.02} \approx 1.122$$

- The euro trades at a forward premium.
- This reflects higher U.S. interest rates.

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FX Forwards vs FX Futures

- FX exposure can be hedged using:
  - FX Forwards (OTC)
    - Bilateral contract
    - Customized maturity and size
    - Settled at maturity
    - Counterparty credit risk
  - FX Futures (Exchange-Traded)
    - Standardized contract size and maturity
    - Daily marking-to-market
    - Margin requirements
    - Central clearinghouse
- Pricing Relationship
  - Under deterministic interest rates: $F_0^{futures} = F_0^{forward}$
  - Differences arise when interest rates are stochastic (due to marking-to-market convexity effects).
- Economic Role
  - Lock in exchange rates
  - Implement covered interest parity
  - Convert foreign currency exposure into domestic cash flows

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When Covered Interest Parity Deviates

- CIP predicts: $F_0 = S_0 e^{(r_d - r_f)T}$
- Empirically, since 2007: $F_0 \neq S_0 e^{(r_d - r_f)T}$
- Define cross-currency basis b: $F_0 = S_0 e^{(r_d - r_f + b)T}$
- Persistent basis observed across major currencies.
- This represents a deviation from frictionless arbitrage.

The Arbitrage Puzzle

- Textbook arbitrage:
  - If $b \neq 0$:
    - Borrow in low-rate currency
    - Convert at spot
    - Invest in high-rate currency
    - Hedge via forward
    - Earn riskless profit
  - In frictionless markets: $b \rightarrow 0$
  - Yet since 2007:
    - Basis persists
    - Even when counterparty credit risk is low
- Why does arbitrage not eliminate it?

Theory as to Why the Basis Opens: Hedging Demand

- Bank of International Settlements (2016) identifies structural USD hedging demand from:
  - Global banks with currency funding gaps
  - Institutional investors hedging foreign bond portfolios
  - Corporates issuing foreign-currency debt
- These flows are:
  - Large
  - Persistent
  - Relatively price-insensitive
- Forward price adjusts.
- Basis opens.

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Theory as to Why the Basis Persists: Limits to Arbitrage

- Post-crisis regulatory environment:
  - Basel III capital requirements
  - Leverage ratio constraints
  - Counterparty exposure limits
  - Liquidity coverage requirements
- Arbitrage trades consume:
  - Balance sheet
  - Repo funding capacity
  - Regulatory capital
- “Balance sheet space is rented, not free.”
- Basis reflects the price of balance sheet capacity.

Funding Reality vs Textbook CIP

- Covered interest parity is a frictionless theory.
- It assumes:
  - Unlimited balance sheet capacity
  - Equal borrowing and lending rates
  - No capital or regulatory costs
  - Costless arbitrage
- In practice, arbitrage requires:
  - Funding (often via repo markets)
  - Balance sheet capacity
  - Regulatory capital
  - Counterparty exposure limits
  - These are not free.
- What Changes Post-2008
  - Banks fund positions in secured repo markets
  - Repo rates can diverge across currencies
  - Balance sheet space is scarce
  - Capital requirements constrain arbitrage
- As a result:
  - Arbitrage is no longer costless
  - CIP becomes an approximation
  - The cross-currency basis reflects funding and balance sheet costs

The Secured Overnight Financing Rate (SOFR)

- SOFR is the primary U.S. dollar risk-free benchmark rate.
- It is:
  - The volume-weighted median rate on overnight Treasury repo transactions
  - Based on actual transaction data
  - Published daily by the Federal Reserve
  - Secured by U.S. Treasury collateral
- SOFR reflects the cost of secured overnight borrowing in the U.S. repo market.
- Because it is secured and transaction-based:
  - It contains minimal bank credit risk
  - It is considered a near risk-free reference rate

SOFR and Benchmark Reform

- Historically, most USD derivatives referenced LIBOR:
  - An unsecured interbank borrowing rate
  - Forward-looking term structure
  - Included bank credit and liquidity premium
- Following the financial crisis and benchmark reform:
  - LIBOR was discontinued (USD LIBOR ceased in 2023)
  - SOFR replaced LIBOR as the standard floating benchmark
- Today:
  - New USD swaps reference SOFR
  - Short-term rate futures reference SOFR
  - Term structures are built from SOFR-based instruments

Transition to Interest Rate Futures

- To price interest rate derivatives, we must construct a zero curve.
- We begin with:
  - SOFR futures
  - Forward rate extraction
  - Convexity adjustment
  - Bootstrapping discount factors
- This provides the foundation for:
  - FRAs
  - Swaps
  - Risk-neutral valuation

SOFR Futures: Contract Structure

- SOFR futures are exchange-traded contracts referencing compounded SOFR.
- Two main types:
  - 1-Month SOFR Futures
    - Based on arithmetic average of daily SOFR
    - Reference a calendar month
    - Used primarily for very short-term exposure
  - 3-Month SOFR Futures
    - Based on compounded SOFR
    - Reference Internation Monetary Market quarterly dates
    - Used for forward rate extraction

SOFR Futures: Pricing

- SOFR futures are quoted as: Price=100−R_{ann}
  - where: R_{ann} = annualized rate implied by the contract.
- Extracting the Implied Rate
  - Given futures price P: R_{ann} =100−P
  - Example:
    - If P=95.25
    - Then R_{ann} = 4.75% ->This is an annualized rate.
- Accrual Over the Contract Period
  - If the contract spans D days: $$\text{Accrual over period} = R_{\text{ann}} \times \frac{D}{360}$$
  - (Quoted using ACT/360 convention.)

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Convexity Adjustment

- If interest rates are stochastic:
  - Futures gains occur when rates move
  - Interim cash flows are reinvested
  - Correlation between rate level and discount factors matters
- This produces a convexity effect.
- Typically: Futures Rate>Forward Rate
- Magnitude depends on:
  - Interest rate volatility
  - Time to maturity
- For short maturities (e.g., 1–2 years):
  - Adjustment is small
  - Often ignored in introductory construction
- We denote the forward rate: $f_{t1,t2} = R_{ann}$−Convexity Adjustment

Constructing Discount Factors (First Segment)

- Suppose convexity-adjusted forward rate for first quarter is:
  - $f_{0,0.25}$=4.80%
- Accrual fraction:
  - Δ=0.25
- Discount factor:

$$P(0, 0.25) = \frac{1}{1 + f_{0,0.25}\Delta}$$

- This gives the first point on the SOFR discount curve.

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Bootstrapping the Curve (Numerical Example)

- From previous slide: $f_{0,0.25} = 4.80\%, \quad \Delta = 0.25$
- Step 1 – First Discount Factor:

$$P(0, 0.25) = \frac{1}{1 + 0.048 \times 0.25}$$

$$P(0, 0.25) = \frac{1}{1.012} \approx 0.9881$$

- Now suppose: $f_{0.25,0.50} = 5.00\%$
- Step 2 – First Discount Factor:

$$P(0.25, 0.50) = \frac{1}{1 + 0.05 \times 0.25}$$

$$P(0.25, 0.50) = \frac{1}{1.0125} \approx 0.9877$$

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Bootstrapping the Curve (Numerical Example) (con.)

- Step 3: Chain Discount Factors:

$$P(0, 0.50) = P(0, 0.25) \times P(0.25, 0.50)$$

$$P(0, 0.50) = 0.9881 \times 0.9877$$

$$P(0, 0.50) \approx 0.9759$$

- We know have:

$$P(0, 0.25) \approx 0.9881$$

$$P(0, 0.50) \approx 0.9759$$

- This is the first portion of the SOFR discount curve.

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Extending the Curve: Overnight Indexed Swaps (OIS)

- SOFR futures provide forward rates at the short end of the curve.
- To extend the curve beyond the liquid futures horizon, markets use SOFR OIS swaps.
- An Overnight Indexed Swap (OIS):
  - Fixed leg pays fixed rate K
  - Floating leg pays compounded overnight SOFR
  - No exchange of principal
  - Typically collateralized
- Maturities available:1 year, 2 years,5 years,10 years, 30 years
- OIS swaps are the primary instruments used to anchor the long end of the SOFR zero curve.

Structure of a SOFR OIS

- Consider a T-year OIS with payment dates $t_1, t_2, \ldots, t_n$.
- Fixed Leg
  - Pays: $K\Delta_i$ at each payment date t_i.
  - Present value: $\mathrm{PV}_{fixed} = K \sum_{i=1}^{n} P(0, t_i)\Delta_i$
- Floating Leg
  - Pays compounded SOFR over each accrual period.
  - Under no-arbitrage: $\mathrm{PV}_{floating} = 1 - P(0, T)$

Bootstrapping with OIS

- After initiation of a par OIS: $\mathrm{PV}_{fixed} = \mathrm{PV}_{floating}$

- Therefore: $K \sum_{i=1}^{n} P(0, t_i)\Delta_i = 1 - P(0, T)$

- If all earlier discount factors are known:
  - This equation can be solved for the unknown $P(0,T)$
- Thus, OIS rates allow us to sequentially bootstrap discount factors beyond the futures horizon.

The Complete SOFR Zero Curve

- The full risk-free curve is constructed using:
  1. Overnight SOFR (shortest maturity)
  2. SOFR futures (near-term forward segments)
  3. SOFR OIS swaps (longer maturities)
- Bootstrapping yields:
  - Discount factors $P(0,T)$
  - Zero rates $R(0,T)$
  - Forward rates $f_{t1,t2}$
- This SOFR curve is used to price:
  - FRAs
  - Interest rate swaps
  - Options on rates
  - Other USD derivatives
- In modern markets, the OIS curve defines the risk-free discount curve.

Forward Rate Agreements (FRAs)

- A Forward Rate Agreement (FRA) is a contract that locks in an interest rate for a future borrowing or lending period.
- At time 0:
  - Parties agree on rate K
  - For borrowing/lending between t_1 and t_2
  - Accrual fraction Δ
- At time t_1: Settlement occurs based on realized rate $L(t_1)$
- FRA Payoff at t_2 : Payoff=$(L(t_1) - K)\Delta \times N$
  - where:
    - N = notional
    - $L(t_1)$ = reference rate set at t_1
- Often settled at t_1 in discounted form.
- An FRA is a single-period interest rate forward contract.
- Its fair rate is the forward rate implied by the zero curve.

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Pricing an FRA

- No-arbitrage implies: $1 + f_{t_1,t_2}\Delta = \dfrac{P(0,t_1)}{P(0,t_2)}$
- Thus: $f_{t_1,t_2} = \dfrac{1}{\Delta}\left(\dfrac{P(0,t_1)}{P(0,t_2)} - 1\right)$
- This is the forward rate implied by the SOFR zero curve.
- Value of an FRA at Time 0: $\text{Value} = N\Delta(f_{t_1,t_2} - K)P(0,t_2)$
  - If $K = f_{t_1,t_2}$: Value = 0

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Interest Rate Swaps

- What is a swap?
  - An interest rate swap is a bilateral OTC derivative in which two parties exchange cash flows based on a notional principal.
  - Typically, one party pays a fixed rate and the other pays a floating rate (e.g., SOFR).
  - The notional is not exchanged; it is only used to calculate payments.
- Mechanics
  - At each payment date, both legs are calculated on the same notional.
  - Cash flows are netted, and only the difference is exchanged.
  - The floating leg resets periodically to current market rates.
  - The swap's market value fluctuates as interest rates move.
- Why trade swaps?
  - Transform an asset: e.g., hold a floating-rate asset and receive fixed via a swap to lock in fixed income.
  - Transform a liability: e.g., issue fixed-rate debt and enter a pay-fixed swap to synthetically create floating-rate funding
  - Hedge duration or manage interest rate exposure without buying/selling bonds.
  - Express macro or relative-value views on rates.
- How swaps are organized and traded
  - Swaps are traded over-the-counter between dealers and clients.
  - Most are cleared through central counterparties, reducing counterparty risk.
  - Standard maturities and conventions enhance liquidity (e.g., 2y, 5y, 10y).
  - Dealers warehouse risk and hedge dynamically in underlying bond and futures markets.

Interest Rate Swap: Structure

- An interest rate swap exchanges:
  - Fixed rate K
  - For floating SOFR
  - Notional N.
  - Payment dates: $t_1, t_2, \ldots, t_n$
  - No exchange of principal.

- Present Value of Fixed Leg: $\mathrm{PV}_{fixed} = NK \sum_{i=1}^{n} P(0, t_i)\Delta_i$

- Per Unit Notional: $\mathrm{PV}_{fixed} = K \sum_{i=1}^{n} P(0, t_i)\Delta_i$

  - To simplify algebra, we often divide all cash flows by N.
  - "Per unit notional" means: Value per dollar of notional=Total Value divided by N

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Valuing the Floating Leg (Per Unit Notional)

- Each floating coupon equals: $f_{t_{i-1},t_i}\Delta_i$
- Present value of coupon at t_i: $f_{t_{i-1},t_i}\Delta_i \cdot P(0,t_i)$
- Using forward-rate identity: $1 + f_{t_{i-1},t_i}\Delta_i = \dfrac{P(0,t_{i-1})}{P(0,t_i)}$
- Rearranging: $f_{t_{i-1},t_i}\Delta_i = \dfrac{P(0,t_{i-1}) - P(0,t_i)}{P(0,t_i)}$
- Thus, PV of coupon: $P(0,t_{i-1}) - P(0,t_i)$
- Summing across all periods: $\sum_{i=1}^{n}(P(0,t_{i-1}) - P(0,t_i)) = 1 - P(0,T)$
- Therefore: $\mathrm{PV}_{floating} = N(1 - P(0,T))$
- Per unit notional: $1 - P(0,T)$

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Deriving the Par Swap Rate

- At initiation: $\mathrm{PV}_{fixed} = \mathrm{PV}_{floating}$
- Per unit notional: $K \sum_{i=1}^{n} P(0, t_i)\Delta_i = 1 - P(0, T)$
- Solve for K: $K = \dfrac{1 - P(0, T)}{\sum_{i=1}^{n} P(0, t_i)\Delta_i}$
- For notional N, multiply both sides by N.

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Numerical Example: Par Swap Rate

- Suppose the SOFR zero curve implies the following discount factors:
  - $P(0,0.5)=0.9759$
  - $P(0,1.0)=0.9510$
- Assume:
  - 1-year swap
  - Semiannual payments
  - Accrual fraction $\Delta=0.5$
  - Notional $N=100$
- Step 1 — Compute Denominator:

$$\sum P(0,t_i)\Delta_i = 0.9759(0.5) + 0.9510(0.5)$$

$$= 0.48795 + 0.47550$$

$$= 0.96345$$

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Numerical Example: Par Swap Rate (con.)

- Step 2 — Compute Numerator:

$$1 - P(0,T) = 1 - 0.9510 = 0.0490$$

- Step 3 — Par Swap Rate:

$$K = \frac{0.0490}{0.96345} = 0.0509$$

$$K \approx 5.09\%$$

- This is the fixed rate that makes the swap value zero at initiation.

Mark-to-Market of an Existing Swap

- Suppose:
  - You entered the swap at fixed rate K=5.09%
  - Notional N=100
  - Later, the zero curve shifts and new discount factors are:
    - $P'(0,0.5)=0.9700$
    - $P'(0,1.0)=0.9400$
- Step 1 — Compute New Par Swap Rate
  - Denominator: 0.9700(0.5)+0.9400(0.5) = 0.9550
  - Numerator: 1−0.9400 = 0.0600
  - New swap rate: $K' = \frac{0.0600}{0.9550} = 6.28\%$

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Mark-to-Market of an Existing Swap (con.)

- Value of Original Swap:
  - Original fixed rate = 5.09%
  - Market rate = 6.28%
  - Per unit notional: $$\text{Value} = (K - K') \sum P'(0, t_i)\Delta_i$$
    $$= (0.0509 - 0.0628)(0.9550) = -0.0114$$
  - Multiply by notional N=100: Swap Value ≈ −1.14
- Rates increased.
- Fixed rate you pay is below market.
- Swap has negative value to fixed-rate payer.
- Swaps are marked to market using updated discount factors.

DV01 and Duration of Interest Rate Swaps

- Consider a fixed-for-floating swap with:
  - Notional: N
  - Fixed rate: K
  - Discount factors: $P(0,t_i)$
  - Accrual fractions: Δi
- Recall: $\mathrm{PV}_{swap} = N\left[K\sum P(0,t_i)\Delta_i - (1 - P(0,T))\right]$ (for a fixed rate payer)
- DV01 (Dollar Value of a 1 bp Move)
  - DV01 measures the change in swap value for a 1 basis point parallel shift in rates: $\mathrm{DV01} = \frac{\partial \mathrm{PV}}{\partial y} \times 0.0001$
  - For a fixed-rate payer, approximately: $\mathrm{DV01} \approx N\sum P(0,t_i)\Delta_i \times 0.0001$
  - DV01 equals the present value of a 1 bp change in the fixed leg.
  - Floating leg has minimal first-order sensitivity at initiation.

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DV01 and Duration of Interest Rate Swaps (con.)

- DV01 and Duration of Interest Rate Swaps
  - At par: $\mathrm{PV}_{floating} = 1 - P(0,T)$
  - Small parallel rate shifts:
    - Affect discount factors
    - But floating coupons reset to market rates
  - Thus: Swap duration ≈ fixed-leg duration.
- Swap Duration
  - Modified duration of a swap: $D_{swap} = \dfrac{\mathrm{DV01}}{\mathrm{PV}_{swap}}$
  - At initiation (par swap): $D_{swap} \approx \dfrac{\sum P(0,t_i)\Delta_i}{1 - P(0,T)}$

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DV01 and Duration of Interest Rate Swaps (con.)

- A fixed-rate payer swap behaves like:
  - Short a fixed-rate bond
  - Long a floating-rate note
- Net duration is close to:
  - Duration of fixed leg
  - Floating leg duration ≈ very small
- Swaps are therefore powerful tools for:
  - Adjusting portfolio duration
  - Hedging interest rate exposure
  - Managing yield curve risk

Numerical Example: Swap DV01 (PVBP)

- Use the same 1-year, semiannual swap structure from the par-swap example.
  - Assume discount factors (today):
    - P(0,0.5)=0.9759
    - P(0,1.0)=0.9510
    - Δ1=Δ2=0.5
- Step 1 — Compute the "Fixed-Leg Annuity"

$$A = \sum_{i=1}^{2} P(0, t_i)\Delta_i = 0.9759(0.5) + 0.9510(0.5)$$

$$A = 0.48795 + 0.47550 = 0.96345$$

- This is the PV of receiving $1 per year of fixed coupon payments over the swap's payment dates (per $1 notional).

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Numerical Example: Swap DV01 (PVBP)(con.)

- Step 2 — DV01 (PVBP)
  - DV01 = change in swap PV for a 1 bp (= 0.0001) change in the fixed rate, holding discount factors fixed (the standard "PVBP of the fixed leg" approximation):

$$\mathrm{DV01} \approx N \cdot A \cdot 0.0001$$

  - Assume Notional N=$100,000,000:

$$\mathrm{DV01} \approx 100{,}000{,}000 \times 0.96345 \times 0.0001 = 9{,}634.5$$

  - So, DV01 is approximately $9,635 per bp

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Currency Swaps

- A currency swap exchanges:
  - Principal in currency A
  - For principal in currency B
  - With periodic interest payments
  - Final re-exchange of principal at maturity
- Usually, the principal amounts are chosen to be approximately equivalent using the exchange rate at the swap's initiation.
- But when they are exchanged at the end of the life of the swap, their values may be quite different.
- Main Types:
  - Fixed-for-Fixed Currency Swap
  - Fixed-for-Floating Currency Swap
  - Floating-for-Floating (Cross-Currency Basis Swap)
    - Floating in both currencies
    - Includes basis spread
- Currency swaps differ from interest rate swaps because:
  - Principal is exchanged.
  - Exchange rate risk is central.

Comparative Advantage Theory

- Early explanation of currency swaps:
  - Firms have different borrowing advantages across currencies.
  - Each borrows where it has comparative advantage.
  - Swap redistributes exposure.
- Example:
  - U.S. firm borrows cheaply in USD.
  - European firm borrows cheaply in EUR.
  - Swap allows each to obtain desired currency at lower cost.
- Modern Interpretation
  - Today, currency swaps primarily reflect:
  - Funding conditions
  - Regulatory constraints
  - Balance sheet costs
  - Cross-currency basis

Valuing a Fixed-for-Fixed Currency Swap

- Consider a USD/EUR fixed-for-fixed swap.
  - At initiation, exchange N_{USD} for N_{EUR} at spot S_0.
  - Over life, pay fixed USD coupons and receive fixed EUR coupons.
  - At maturity, re-exchange principal.
- Here, we value from the perspective of the USD payer / EUR receiver.
- Valuation Approach 1 — Forward Conversion at Each Date
  - For a foreign cash flow $C_f(t_i)$:
  - Step 1: Convert at the forward rate for that maturity:

$$C_d(t_i) = F(0, t_i)\, C_f(t_i)$$

  - Step 2: Discount domestically:

$$\mathrm{PV} = P_d(0, t_i)\, F(0, t_i)\, C_f(t_i)$$

Valuing a Fixed-for-Fixed Currency Swap (con.)

- Valuation Approach 1 (con.)
  - Use Covered Interest Parity (CIP): $F(0,t_i) = S_0 \frac{P_f(0,t_i)}{P_d(0,t_i)}$
  - Substitute: $\mathrm{PV} = P_d(0,t_i) \cdot S_0 \frac{P_f(0,t_i)}{P_d(0,t_i)} \cdot C_f(t_i)$
  - Rearrange: $= S_0\, P_f(0,t_i)\, C_f(t_i)$
  - Result
    - Domestic PV of foreign cash flows = $S_0 \times$ Foreign PV

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Valuing a Fixed-for-Fixed Currency Swap (con.)

- Valuation Approach 2 — Bond Representation
  - A fixed-for-fixed currency swap is equivalent to:
    - Long a foreign fixed-rate bond
    - Short a domestic fixed-rate bond
  - Thus: $V = S_0 B_{foreign} - B_{domestic}$
  - Where:

$$B_{foreign} = \sum P_f(0, t_i) C_f(t_i) + P_f(0, T) N_f$$

$$B_{domestic} = \sum P_d(0, t_i) C_d(t_i) + P_d(0, T) N_d$$

  - Under covered interest parity Forward-based valuation and Bond-price valuation are identical.
  - No expectation of future spot rates is required.
  - Currency swap valuation depends only on:
    - Spot rate S_0
    - Domestic discount curve
    - Foreign discount curve

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Variations on Interest Rate Swaps

- Swaps can be structured in many ways to tailor cash flows and risk.
- Common structures:
  - Amortizing swap — notional declines over time
  - Accreting swap — notional increases over time
  - Forward-start swap — begins at a future date
  - Step-up/step-down swap — fixed rate changes over time
  - Zero-coupon swap — fixed leg paid only at maturity
- These alter:
  - Cash flow timing
  - Notional profile
  - Duration exposure

Swaps with Embedded Options

- Swaps can be structured in many ways to tailor cash flows and risk.
- Examples:
  - Callable swap — one party can terminate early
  - Putable swap — counterparty can terminate
  - Extendable swap — maturity can be lengthened
  - Cancelable swap — early termination right
- These introduce:
  - Optionality
  - Convexity
  - Volatility exposure
- Valuation requires:
  - Lattice methods
  - Monte Carlo simulation
  - Short-rate models

Equity Swaps

- Equity swaps exchange:
  - Fixed or floating rate
  - For total return on an equity index or stock
- Total return includes:
  - Price appreciation
  - Dividends
- Used for:
  - Synthetic equity exposure
  - Leverage
  - Balance sheet management

Quanto Swaps

- A quanto provides exposure to:
  - A foreign asset
  - With payoff settled in domestic currency
- The investor gains asset exposure without direct FX exposure.
- Pricing reflects correlation between:
  - Asset returns
  - Exchange rates

Commodity Swaps

- Commodity swaps exchange:
  - Fixed price
  - For floating commodity price
- Used by:
  - Producers to hedge output prices
  - Airlines to hedge fuel costs
  - Industrial firms to stabilize input expenses
- Closely linked to:
  - Futures term structure
  - Storage and convenience yield

Volatility and Variance Swaps

- Volatility swap:
  - Pay fixed volatility
  - Receive realized volatility
- Variance swap:
  - Pay fixed variance
  - Receive realized variance
- Key property:
  - Variance swaps can be replicated using options across strikes.
  - They provide pure volatility exposure.
- Used for:
  - Tail-risk positioning
  - Hedging options portfolios
  - Volatility plays

Credit Default Swaps (CDS)

- A Credit Default Swap is a contract that transfers credit risk.
- It functions much like an insurance contract on debt.
- Structure
  - Buyer of protection pays periodic premium (CDS spread)
  - Seller of protection compensates buyer if default occurs
  - Reference entity = company or sovereign being insured
- A CDS allows investors to:
  - Hedge credit exposure
  - Take directional views on credit risk
  - Isolate default risk from interest rate risk
- CDS spreads reflect:
  - Expected default losses
  - Risk premia
  - Market liquidity
- CDS transformed credit risk into a tradable asset class.